

Vamsi Krishna Sistla

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EDUCATION

National Institute of Technology, Jamshedpur

Sep 2023 – Jun 2027

B.Tech in Electrical Engineering | CGPA: 8.36

Narayana Junior College

Apr 2021 – May 2023

Intermediate (MPC) | Grade: 93.2%

KKR Gowtham School

Apr 2020 – May 2021

Matriculation | Grade: 89%

RELEVANT PROJECTS

SMPS Mobile Charger Design (Power Electronics)

- Designed a compact 5V/2A Switched Mode Power Supply (SMPS) mobile charger using Flyback converter topology.
- Calculated and designed the high-frequency ferrite core transformer and implemented feedback control using an Optocoupler and TL431 for stable voltage regulation.
- Performed circuit simulation to analyze line and load regulation, followed by component selection and hardware soldering on a general-purpose PCB.
- *Tech Stack: Power Electronics, PCB Design, Transformer Design, PWM Control*

Simulation of Battery-Connected Solar Power System (MATLAB Simulink)

- Simulated a standalone Solar PV System with integrated battery energy storage using MATLAB Simulink.
- Implemented Maximum Power Point Tracking (MPPT) control logic to optimize solar energy harvest.
- Developed charging and discharging logic for the battery storage system to ensure reliable power supply and grid stability.
- *Tech Stack: MATLAB, Simulink*

TECHNICAL EXPERTISE & SKILLS

- **Core EE Domains:** Electrical Machines, Power Systems, Power Electronics, Control Systems
- **Simulation & Tools:** MATLAB Simulink, Arduino IDE, Git/GitHub
- **Programming Languages:** C/C++, Python, SQL, JavaScript
- **CS Fundamentals:** Data Structures And Algorithms, OOPS, DBMS
- **Domains:** web development, Data science

POSITIONS OF RESPONSIBILITY

- **Active Member, Electrical Engineering Society (EES), NIT Jamshedpur:** Active contributor to the departmental society, organizing events and technical workshops relevant to electrical engineering.
- **Active Member, Official Coding Club (PCON), NIT Jamshedpur:** Participated in organizing coding events and fostering a programming culture within the institute.

ACHIEVEMENTS & CERTIFICATIONS

- **Rank 4 in CODE CLASH**, a coding competition by Electrical Engineering Society (EES), NIT Jamshedpur.
- **Solved 500+ competitive programming problems** across HackerRank, Codeforces, LeetCode, and other platforms.
- **Machine Learning Specialization** by Andrew Ng (Coursera).